JOSÉ MANUEL REQUENA PLENS

Firmware & Software Engineer
C · C++ · Python · RTOS · Test Automation · CI/CD

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For more information you can click on the name of places, certificates or courses that appear in the curriculum.
Availability: on-site, remote or hybrid.

WORK EXPERIENCE

Software Engineer

Power Electronics

April 2023 – May 2026 · R&D Department. Solar area.

Firmware development for solar inverters and control systems in photovoltaic plants, working on STM32 microcontrollers with bare-metal and RTOS environments.

- Reduced RTOS task stack usage by 30%–50%, optimizing memory on STM32 microcontrollers.
- Optimized Modbus TCP request handling, cutting CPU usage under load (40,000 req/s) from 80% to 17% with no dropped responses.
- Redesigned embedded file-system access, cutting access times by over 50% and adding a scheduled purge (daily/monthly) that safely clears thousands of files under continuous operation.
- Built and maintained industrial communications in C with hardware abstraction layers (HAL): Modbus RTU/TCP, UART, I2C and SPI.
- Automated the test suite: ∼50–80 unit tests per module (Ceedling, thousands in total) reaching 100% line and branch coverage via standard-library *mocks/fakes*, plus hundreds of on-hardware integration tests (pytest); all in CI.
- Reviewed merge requests across a 12-person team, catching bugs and proposing improvements before integration.

Predoctoral researcher

Polytechnic University of Valencia + CSIC + Hospital La Fe + AVI

January 2023 – April 2023 · Medical Imaging and Therapy Systems i3M

Researcher in the R&D line: **Acoustic Metamaterials for Ultrasound Histotripsy treatments**.
Project: New generation of smart metasurfaces based on additive manufacturing for strategic applications in telecommunications (Metasmart). Funded by the Valencian Agency for Innovation (Reference: INNEST/2022/345).

- Ultra-close focus acoustic lens design.
- Prototype manufacturing.
- Experimentation with ex-vivo tissues.

Predoctoral researcher

Polytechnic University of Valencia + CSIC

July 2021 – July 2022 | Medical Imaging and Therapy Systems i3M

The main objective is to conduct research and development of acoustic metamaterials for use in architectural applications and medical imaging.

Goals achieved:

- Developed acoustic metadiffusers reaching a diffusion coefficient of 0.8 with a thickness 10$\times$ smaller than traditional diffusers.
- Simulation of the acoustic diffusers and validation of results (MATLAB & COMSOL).
- Two publications in national and international congresses:
  - Tecniacústica - **Beyond Schroeder diffusers using acoustic metasurfaces**
  - Euronoise - **Sound diffusing metasurfaces based on elastic plates and membranes**
- Designed and 3D-printed (resin) slitted-tube waveguides with numerically-computed slits (2–3 cm tubes, $\sim$0.5 mm slits) that increased ultrasound directivity and range $\sim$15$\times$ in air, validated experimentally.
- Validated experimental results against those obtained in simulation.
- Advisor of the Master thesis of a student of the Master's Degree in Acoustic Engineering of the UPV.

Researcher in European project

Higher Polytechnic School of Gandia

February 2020 – July 2021 | Complex Media Acoustics Group of IGIC

Implement and experimentally validate a sound mitigation method, applicable to a real space vehicle launch configuration (VEGA), that results in a significant decrease in sound pressure levels generated in the launch area during spacecraft liftoff.

Project funded by ESA (European Space Agency) with reference ESA AO/1-9479/18/NL/LvH, in collaboration with: CNRS/Laboratoire d'Acoustique de la Université du Mans; COMET Ingeniería; Spanish National Research Council (CSIC) and Polytechnic University of Madrid.

All project objectives were completed at 100%:

- Designed the optimum acoustic-metamaterial geometry, achieving near-total absorption ($\alpha \approx 1$) across 200–500 Hz with a subwavelength design (ratio λ/thickness ≈ 16.5 vs $\sim$2 conventional), validated on real lab prototypes.
- Simulation of a real environment to validate design performance (MATLAB, COMSOL & Python).
- Designed and 3D-printed the first prototype; for the large-scale version designed the injection-molding model and sourced the manufacture of 600 modules with a supplier.
- Development of software for acoustic measurements according to ISO 10534-2 and ASTM E2611 standards. **A|Lab**.
- Design and development of a 3-axis robotic system to perform experimental measurements. Photos and some information here: **NEWS**.
- Results validated and accepted by ESA.
- Three publications/presentations at international congresses (Euronoise and ECSSMET) and one at a national congress (Tecniacústica):
  - Euronoise - **Perfect broadband sound absorber metamaterial for noise reduction in a rocket launch**
  - Euronoise - **Application of metamaterials to control noise scattering during space vehicle lift-off**
  - ECSSMET - Launch sound level characterisation and mitigation: numerical modelling framework and metamaterial proof of concept
  - Tecniacústica - **Acoustic field prediction during the launch of rockets**

Research internship

Dep. of Physics, Systems Engineering and Signal Theory

February 2018 – July 2018 University of Alicante

Internship in different research projects in the **Applied Acoustics Group**, belonging to the University Institute of Physics Applied to Sciences and Technologies of the Higher Polytechnic School of Alicante.

Goals achieved:

- Simulation of acoustic models.
- Acoustic measurements by means of a LabView-controlled robotic system.
- Investigation of the acoustic radiation efficiency (Vibration-borne noise) of metallic plates for an initial study of ship noise emissions in water. Carried out in collaboration with SAES, here you can read a news item on the subject: **NEWS**.
- A publication at a national congress:
  - Tecniacústica - **Comportamiento vibroacústico de contenedores cilíndricos en aire**

Founder and Vocal Member

AμTech

September 2016 – July 2018 University of Alicante

Association based at the Higher Polytechnic School of the University of Alicante. Founded in 2016. Promotes project-oriented microcontroller programming knowledge.

Activities performed:

- Group classes of reinforcement and extension of the subject ”Digital Electronic Systems” of the Degree in Sound and Image Engineering of the University of Alicante.
- Laboratory organization and planning (located at the Colegio Mayor).
- Communication and Public Relations.

Sound technician

Acusticox

June 2011 – February 2018 Cox, Alicante

Installation and tuning of audio equipment for live events and permanent installations; FOH technician.

OTHER KNOWLEDGE AND SKILLS

Embedded Firmware & Electronics

C	●●●●○
STM32 & ARM	●●●●○
ESP32 / ESP-IDF	●●●●○
PlatformIO	●●●●○
Texas Instruments	●●●○○
FreeRTOS / RTOS	●●●●○
Comm. Protocols	●●●●○
HW Debugging	●●●●○
Keil uVision	●●●●○
Secure Boot & Embedded Sec.	●●●○○
Doxygen	●●●●○
Instrumentation	●●●●○

Quality, CI/CD & DevSecOps

GitHub Actions	●●●●○
GitLab CI	●●●○○
Jenkins	●●○○○
SonarQube / SonarCloud	●●●●○
GoReleaser	●●●○○
Static Analysis & SAST	●●●○○
Testing	●●●●○
pre-commit hooks	●●●●○

Networking

TCP/IP & Networking	●●●●○
MikroTik RouterOS	●●●●○
WireGuard / VPN	●●●●○
IPv6 & Dual-stack	●●●●○
QUIC / HTTP3	●●●○○
DNS / Dynamic DNS	●●●○○

Software Engineering & Tools

Python	●●●●○
C++	●●●○○
Go	●●○○○
TypeScript	●●○○○
Bash	●●●○○
Clang Tooling	●●●○○
Git	●●●●○
GitHub	●●●●○
GitLab	●●●●○
Docker	●●●●○
MATLAB	●●●●○
CMake	●●●○○
JetBrains Toolbox	●●●●○

Security & Cryptography

Applied Cryptography	●●●○○
TLS / mTLS / PKI	●●●●○
Nginx Hardening	●●●●○
CrowdSec	●●●○○
Network Hardening	●●●●○

AI & Agentic Engineering

Model Context Protocol (MCP)	●●●●○
AI Agents & Best Practices	●●●●○
LLM Integration	●●●○○

Simulation & Acoustics

COMSOL Multiphysics	●●●●○
LabView	●●●○○
Acoustic Instrumentation	●●●●○

General & Platforms

Linux / Unix	●●●●○
MacOS	●●●●○
Windows	●●●●○
LaTeX	●●●●○
HTML / CSS	●●●○○

EDUCATION

Formal education

PhD in Health and Wellness Technologies

Polytechnic University of Valencia

September 2020 – 2023 (not completed) — Medical Imaging and Therapy Systems i3M

Title Industrial and biomedical applications of metamaterials for the control of acoustic beams. In this project, we propose to combine new air-coupled ultrasound transducers with hyperbolic metamaterials to design vision systems that exceed the imaging capabilities of traditional systems.

Description Research in acoustic engineering applied to biomedicine: biomedical signal processing, ultrasound characterization, and computational modeling and simulation of acoustic phenomena. I redirected my career toward industrial software development before completing the doctorate, applying my foundation in programming, data analysis and analytical problem-solving.

Academic record

Master's Degree in Acoustical Engineering

Polytechnic University of Valencia

September 2018 – June 2019 — Higher Polytechnic School of Gandia

Degree obtained with honors in: Fundamentals of Acoustics, Acoustic Isolation, Musical Acoustics, Signal Processing in Acoustic Engineering, Ultrasonics and Acoustic Simulation Techniques.

Final Master's thesis entitled "Difusores acústicos basados en resonadores de membrana y placa" ("Acoustic diffusers based on membrane and plate resonators"), qualified with honors with special mention.

Diploma | Academic record

Degree in Telecommunication Engineering in Sound and Image

University of Alicante

July 2018 — Higher Polytechnic School

Degree obtained together with the 2 specializations: Acoustic Engineering and Audiovisual Technology.

Final Degree Work entitled "Estudio de la relación campo directo/reverberado; útil/perjudicial" ("Study of the direct/reverberated field relationship; useful/harmful").

Diploma | Academic record

Certificate of Higher Education in Electronics Technician & Superior Sound Technician

IES Salesianos (San José Artesano) | IES Luis García Berlanga + Ciudad de la Luz

2009 | 2011 — Elche | Alicante

Diplomas

CERTIFICATES

The certificates can be accessed by clicking on the name of each one.

Occupational Health and Safety

Generic	INVASSAT. **50** H.
Nanomaterials	INVASSAT. **50** H.
Chemical sector	INVASSAT. **50** H.
Emergencies	INVASSAT. **70** H.
Nutrition	INVASSAT. **50** H.
Educational	INVASSAT. **50** H.
Services	INVASSAT. **50** H.
Researcher	UPV. **15** H.

Labor/Industrial

Op. industrial trucks	Gescoform. **15** H.
Food Hygiene Course	Asonaman. **30** H.
Self-protection plans	INVASSAT. **15** H.
Static electricity	INVASSAT. **15** H.
Data privacy	CSIRT-CV. **10** H.

Transversal competences

Gender perspective	EVES. **20** H.
Teamwork	Labora. **25** H.
Design Thinking	Labora. **25** H.
Critical thinking	Labora. **25** H.
Adaptability, flexibility	Labora. **25** H.
Autonomy, innovation	Labora. **25** H.
Prof. efficiency	Labora. **25** H.
Entrepreneurship	UPV. **20** H.
Gender persp. in research	UPV. **50** H.

Information technology

Using Python for Research	HarvardX. **50** H.
Analyzing Data w/ Python	IBM. **20** H.
Visualizing Data w/ Python	IBM. **20** H.
Numerical methods (MATLAB)	UPV. **50** H.
Scientific computing	UPV. **50** H.
Documents with LaTeX	UPV. **56** H.